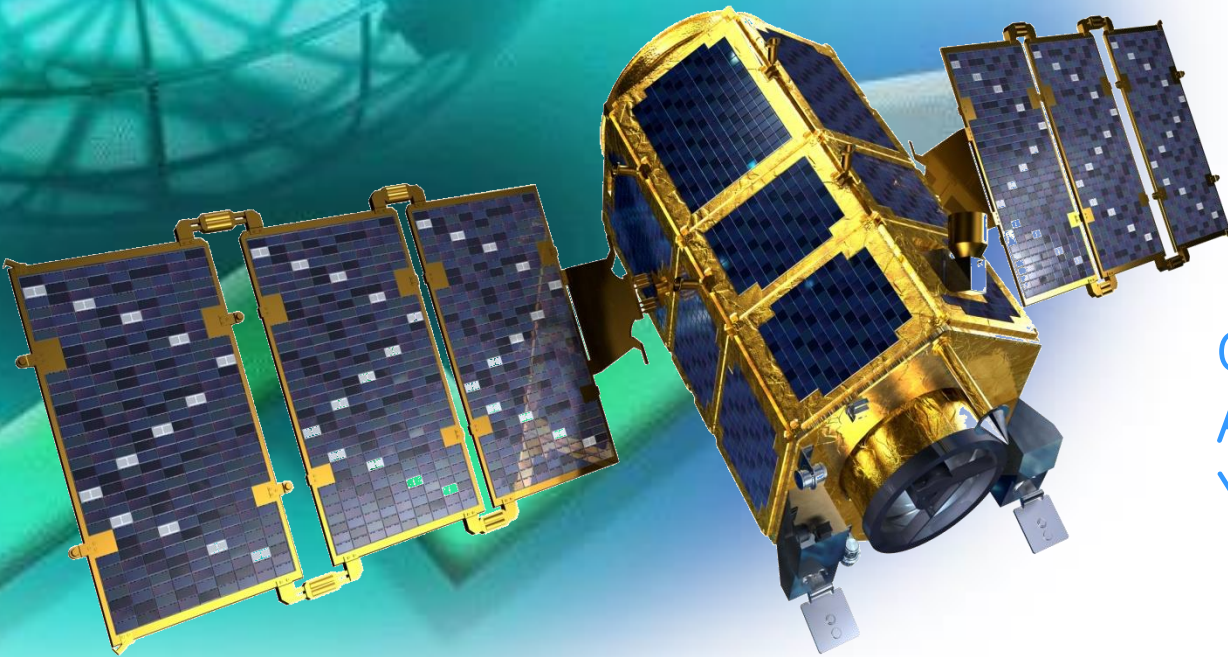
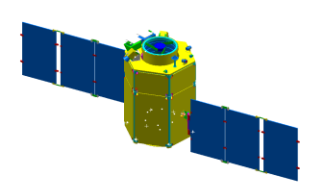


Fall 2011

Spacecraft Systems



Chandeok Park
Astrodynamics & Control Lab.
Yonsei University



Chapter 11

SPACECRAFT SUBSYSTEMS



11.1 Attitude Determination & Control Subsystem

11.2 Telemetry, Tracking & Command Subsystem

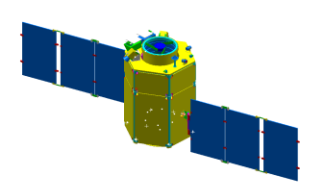
11.3 Command & Data Handling Subsystem

11.4 (Electrical) Power Subsystem

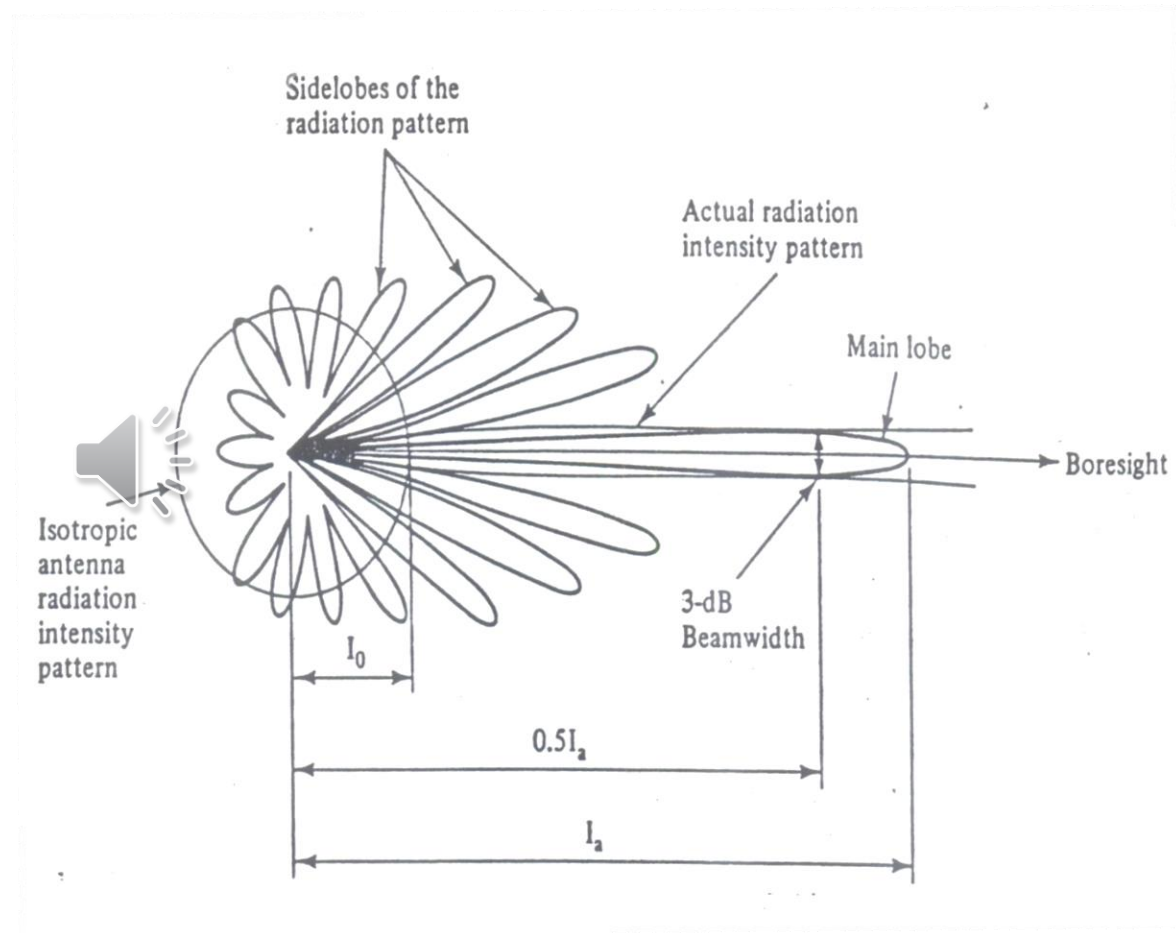
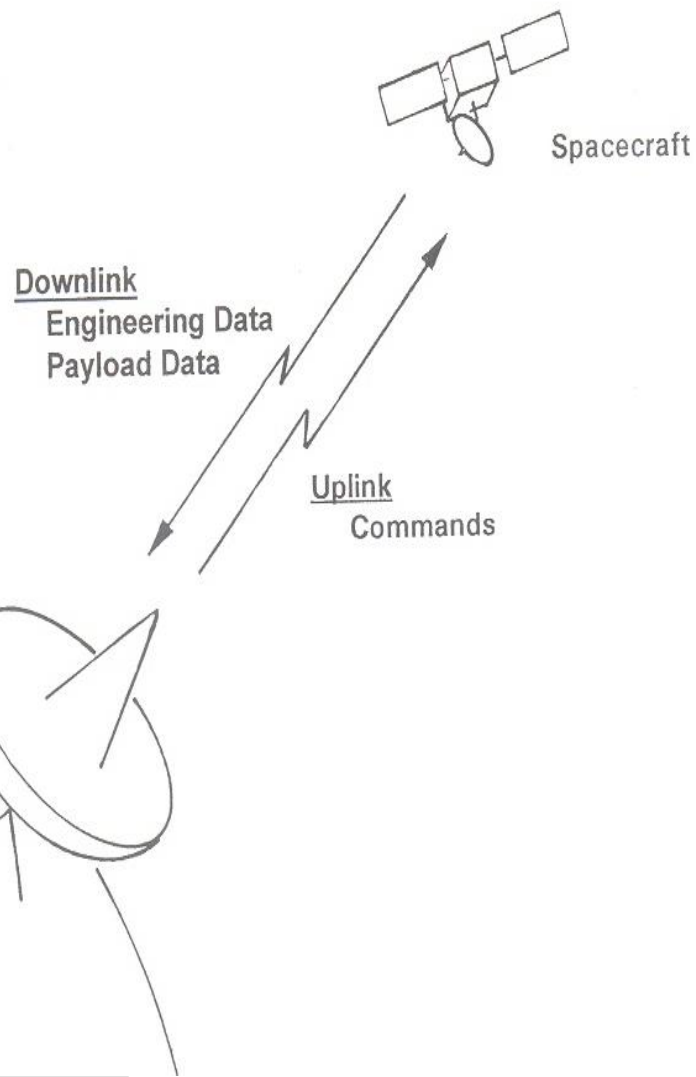
11.5 Thermal Control Subsystem

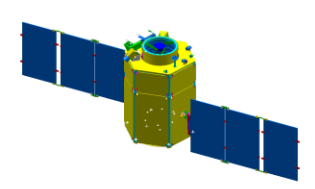
11.6 Structures & Mechanism Subsystem

11.7 Guidance, Navigation & Control Subsystem



11.2 Telemetry, Tracking & Command Subsystem





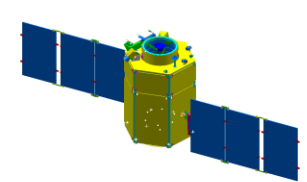
11.2 TT&C

- ◆ Antenna Peak Gain(dB) = $10 \log_{10}(I_a/I_o)$
 - I_a : maximum radiated power intensity
 - I_o : power intensity of an isotropic antenna
- ◆ The main beam of an antenna is the main lobe of the radiation pattern centered around the boresight of the antenna
- ◆ For an efficient antenna, the main beam peak gain should be as high as possible with respect to the isotropic gain, and the sidelobe peaks should be as low as possible

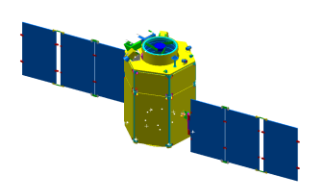
- ◆ Payload mission data and S/C housekeeping (engineering) data pass from S/C through TT&C subsystem to operators (ground system) and users

TABLE 11-18. What a TT&C Subsystem Does. These are the functions of a communications subsystem connecting the satellite to the ground or other satellites. In a broad sense the communications subsystem receives signals from Earth or another satellite and transmits signals to Earth or another satellite.

Specific Functions
• Carrier Tracking (lock onto the ground station signal) – 2-way coherent communication (downlink frequency is a ratio of the uplink frequency) – 2-way noncoherent communication – 1-way communication • Command Reception and Detection (receive the uplink signal and process it) – Acquire and track uplink carrier – Demodulate carrier and subcarrier – Derive bit timing and detect data bits – Resolve data-phase ambiguity if it exists – Forward command data, clock, and in-lock indicator to the subsystem for command and data handling



- **Telemetry Modulation and Transmission** (accept data from S/C systems, process & transmit them)
 - Receive telemetry data streams from the command and data handling subsystem or data storage subsystem
 - Modulate downlink subcarrier and carrier with mission or science telemetry
 - Transmit composite signal to the ground station or relay satellite
- **Ranging** (receive, process, and transmit ranging signals to determine S/C's position)
 - Detect and retransmit ranging pseudorandom code or ranging tone signals
 - Retransmit either phase coherently or noncoherently
- **Subsystem Operations** (process subsystem data, maintain its own health & status, and point the antennas)
 - Receive commands from the subsystem for command and data handling
 - Provide health and status telemetry to the C&DH subsystem
 - Perform antenna pointing for any antenna requiring beam steering
 - Perform mission sequence operations per stored software sequence
 - Autonomously select omni-antenna when spacecraft attitude is lost
 - Autonomously detect faults and recover communications using stored software sequence



◆ 11.2.1 Requirements

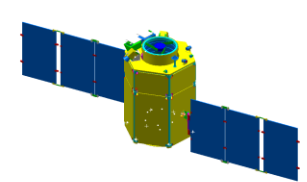
- TT&C subsystem derives its requirements from:
 - Mission or Science Objective
 - Top Level Requirements such as architecture, orbit lifetime and environment)
 - Satellite
 - System Level
 - TT&C Subsystem
 - Internal Requirements
 - Other Satellite Subsystem
 - Ground Station and Any Relay Satellite
 - Compatibility
 - Mission Operations
 - Satellite Operational Modes as a Function of Time



◆ 11.2.1 Requirements

TABLE 11-19. TT&C Subsystem Requirements. These are typical system-level requirements that are imposed on the TT&C subsystem. See also Table 11-23. (Courtesy of TRW)

Requirement	Alternative/Considerations	Comments
Data Rates Command Health & status telemetry Mission/ science	4,000 bps typical, 8–64 bps deep spc 8,000 bps is common Low < 32 bps Medium = 32 bps–1 Mbps High > 1Mbps–1 Gbps	Range 2,000–8,000 bps 40–10,000 bps Mission dependent
Data Volume	Record data, compress data, and transmit during longer windows	Data rate × transmission duration • Shorter duration increases data rate • May require data compression
Data Storage	Solid-state recorders 128×10^6 bits	Policy may dictate all data be stored that is not immediately transmitted Mission may require that data be stored then played back later
Frequency	Use existing assigned frequencies and channels Use systems that are compatible to the existing system	Policy set by FCC, ITU, National Telecommunications & Info. Admin. Refer to the atmospheric frequency absorption charts
Bandwidths	Use C.E. Shannon's theorem to calculate channel capacity; See Chap. 13, Eq. (13-24).	Primary driver is data rate Secondary driver is modulation scheme



◆ 11.2.1 Requirements (cont'd)

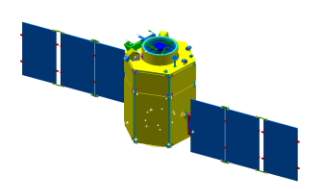
Power to year	Use larger antennas, higher efficiency amplifiers Reconsider data requirements	S/C power may limit size of TT&C system transmitter
Mass to year	Use TWTAs for higher RF power output to reduce antenna size, reconsider data requirements	S/C TT&C system mass allocation may limit size of antennas
Beamwidth to year	See Tables 13-14, 13-15, and 13-16 for various antenna types, beam shapes, and beamwidths 	Ground coverage area requirements or the radiation footprint on the ground Antenna gain null requirements Antenna pointing error
EIRP (Effective Isotropic Radiated Power) (Transmitter Req't)	For a constant EIRP: As antenna size (gain) increases, the transmitter power requirement decreases	$EIRP \text{ (dB)} = \text{transmitter power} +$ $\text{antenna gain} - \text{front end losses}$ Min EIRP required = space loss + atmo loss + antenna pointing loss – receiver antenna gain – receiver sensitivity
G/T [Receiver antenna gain/receiver sys noise temp) (Receiver Req't)	See Table 13-10 for various communication system temperatures and G/Ts	G/T is the sensitivity of the receiving station and a common Figure of Merit; for an existing satellite link, a ground station can only vary its antenna gain and system noise temp to improve the system signal-to-noise ratio

- TWTAs: Travelling Wave Tube Amplifier

◆ 11.2.1 Requirements (cont'd)

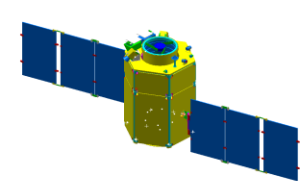
TABLE 11-20. Options for System-Level Interfaces to the TT&C Subsystem. Shown below are several interface possibilities for a TT&C subsystem. If the interface is an existing system, we also provide the system's document number. (Courtesy of TRW)

Interface	Example of Systems	Where to Find Subsystem Parameters
<i>Fixed Ground Station</i>	SGLS—S-band system SDLS—Secure 44/20 GHz GSTDN—S-band 9 and 26 m NASA DSN S-, X-band 26, 34, 70 m Mission-dedicated or unique	TOR-0059 (6110-01)-3, Reissue H MIL-STD-1582 }JPL-DSN-810-5, rev D
<i>Mobile Ground Station</i>	GPS—Ground antenna, S-band, SGLS DMSP MK IV van, S-band AFSCN TMGS—7-m diameter, SGLS-compatible Mission-dedicated or unique • Dedicated comm trailer • Transportable tent	Aerospace Report TOR-0059 (6110-01)-3
<i>Relay Satellite</i>	TDRSS—S-band and Ku-band DSCS III—UHF, S-band, SHF	NASA Goddard STDN 101.2 DSCS III Interface Guide, DCA
<i>Future Relay Sat; Sister Sat w/in Mission Constellation</i>	MILSTAR Mission-dedicated or unique	



◆ 11.2.1 Requirements (cont'd)

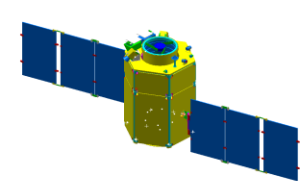
- SGLS: Space–Ground Link System
 - US Air Force
- SDLS: Satellite Data Link Standards
 - US Department of Defense
- GSTDN: Ground Spaceflight Tracking and Data Network
- DSN: Deep Space Network 
 - NASA
- DMSP: Defense Meteorological Satellite Program
 - Department of Defense
- AFSCN: Air Force Satellite Control Network
- TMGS: Transportable/Mobile Ground Station
- TDRSS: Tracking and Data Relay Satellite System
- DSCS: Defense Satellite Communications System
- MILSTAR: Military Strategic and Tactical Relay



◆ 11.2.1 Requirements (cont'd)

TABLE 11-21. TT&C Subsystem Constraints and Requirements on Other Subsystems.
The TT&C subsystem interfaces with all these subsystems and must reliably pass data back and forth or receive support.

Subsystem	Requirement	Constraint
<i>Attitude Determination and Control</i>	• Antenna pointing requirements for gimbaled antennas (gimbal degrees of freedom, amount of rotation) • Pointing requirements of the lesser of 1/10 of antenna beamwidth or 0.3 deg • Closed-loop pointing requirements (i.e., cross-links requiring autotracking)	• Spacecraft pointing and attitude knowledge for fixed antennas may impact antenna beamwidth requirements • Uncertainty in attitude and pointing estimates for the pointing loss in the link budget
<i>Command and Data Handling</i>	• Command and telemetry data rates • Clock, bit sync, and timing requirements • 2-way comm requirements • Autonomous fault detection and recovery requirements (ROM stored command sequence that automatically selects the backup receiver and omni-antenna) • Command & telemetry electrical interface	• Onboard storage and processing



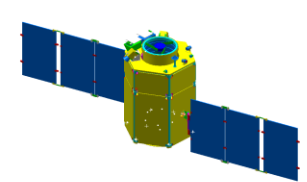
◆ 11.2.1 Requirements (cont'd)

<i>Electrical Power Subsystem</i>	• Distribution requirements	• Amount and quality of power, including requirements for duty cycle, average, and peak power
<i>Structure/ Thermal</i>	• Heat sinks for traveling wave tube amplifiers • Heat dissipation of all active boxes • Location of TT&C subsystem electronics and antennas (locate comm electronics as close to the antennas as possible to minimize RF cable loss) • A clear field of view and movement for all gimbaled antennas	• Temperature uncertainty on non-oven-controlled frequency sources resulting in frequency uncertainty
<i>Payload</i>	• Requirements for storing mission data • RF and EMC interface requirements (conducted emissions, conducted susceptibility, radiated emissions, radiated susceptibility) • Special requirements for modulation, coding, and decoding	• Maximum data rates for mission or science telemetry • Maximum data volume for mission or science telemetry
<i>Propulsion</i>	• None	• None

◆ 11.2.1 Requirements (cont'd)

TABLE 11-22. Preliminary Design Process for the TT&C Subsystem.

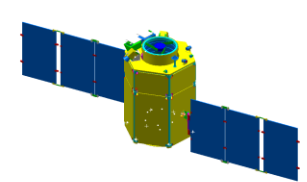
Step	Comments	Where Discussed
1. Determine requirements	Range, orbit and spacecraft geometry Data rate and volume Minimum elevation angle Worst case rain conditions Bit error rate	Table 11-19
2. Select frequency	Typically an existing, assigned frequency	Sec. 11.2.2, Table 13-12
3. Determine required bandwidth	Use Shannon's theorem; primary driver is data rate	Eq. 13-24 (Chap. 13)
4. Do major subsystem trades between: – Receiver noise temperature – Receiver gain (antenna aperture) – Transmitter gain (antenna aperture) – Transmitter power	Use link budget to trade between components	Table 13-13
5. Do major subsystem trades between the TT&C subsystem and other subsystems	Understand the satellite's sensitivity to each TT&C subsystem design feature	Table 11-21
6. Calculate performance parameters	EIRP, G/T, and margin	Tables 11-19, 13-12
7. Estimate subsystem weight and power	Use analogy with existing systems	Table 11-26
8. Document reasons for selection	Important to document assumptions	N/A



◆ 11.2.2 Designing the TTC Subsystem

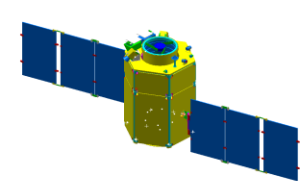
TABLE 11-23. Design Parameters for the TT&C Subsystem. Below are design parameters for the TT&C subsystem that are not typically specified at the system level. (See Table 11-19 for system-level parameters.)

Parameter	Comment
<i>Antenna Sidelobe Levels</i>	Design to minimize. Sidelobes degrade the antenna's directionality. Very high sidelobes may interfere with other antennas and receivers on the satellite. High sidelobe levels also affect security by making detection of signals more likely.
<i>Polarization</i>	Polarizations can be circular (right or left), or linear (horizontal or vertical). To decrease signal loss in the link, the polarizations need to be compatible. For example, the satellite antenna and ground station must both have right-circular polarization.
<i>Frequency Stability</i>	When we need to acquire the signal quickly, the receiver frequency must be known and stable. Thus, we specify the original receiver frequency's set point, short-term stability, temperature stability, and aging stability so we can acquire the uplink signal with little uncertainty.



◆ 11.2.2 Designing the TTC Subsystem (cont'd)

<i>Capture and Tracking Range</i>	The <i>capture range</i> is the band of frequencies over which the uplink-carrier signal can drift from the receiver's best-lock frequency, so the receiver will still lock to the uplink signal. The <i>tracking range</i> is the band of frequencies the receiver will follow while locked to a sweeping, uplink-carrier signal without losing lock. Typically the capture range is 1% of the tracking range.
<i>Diplexer Isolation</i>	The <i>diplexer</i> allows us to use the same antenna for transmitting and receiving. The diplexer isolates the transmitter from the receiver. A diplexer with low isolation may require a band-reject filter between the transmitter and the diplexer.
<i>Coupling Between Antennas</i>	Signal and noise coupling between a transmitting antenna and a receiving antenna may cause the receiving antenna to lock onto a frequency coming from the transmitting antenna's transmitter. More commonly, broadband noise from the transmitting antenna may couple over to the receiving antenna and raise the noise floor of the receiving antenna's receiver, reducing the signal-to-noise ratio.



◆ 11.2.2 Designing the TTC Subsystem (cont'd)

TABLE 11-24. Selection Criteria for the TT&C Subsystem. Selection criteria at the subsystem level fall into three broad categories: performance, compatibility, and other. Chapter 13, Table 13-1, shows the selection process for communications architectures. (Courtesy of TRW)

Category	Criteria	Comments
<i>Performance</i>	• Mass • Volume • Power (RF and dc) • Bit error rate—Figure of Merit • Noise figure—Figure of Merit • Frequency stability • Insertion loss • Reliability • Efficiency	} See Table 11-26 Uplink = 10^{-6}, downlink = 10^{-5} are typical values (Sec. 13.3) 4 dB typical for SGLS Typically in parts per million Measured in terms of mean time between failures (Sec. 19.2) Percent transponder-radiated power/input power
<i>Compatibility</i>	• Compatibility with existing systems • SGLS compatibility • TDRSS compatibility	
<i>Other</i>	• Technology risk • Heritage	Subjective (Sec. 20.4) Measured in terms of previous spaceflight experience

◆ 11.2.2 Designing the TTC Subsystem (cont'd)

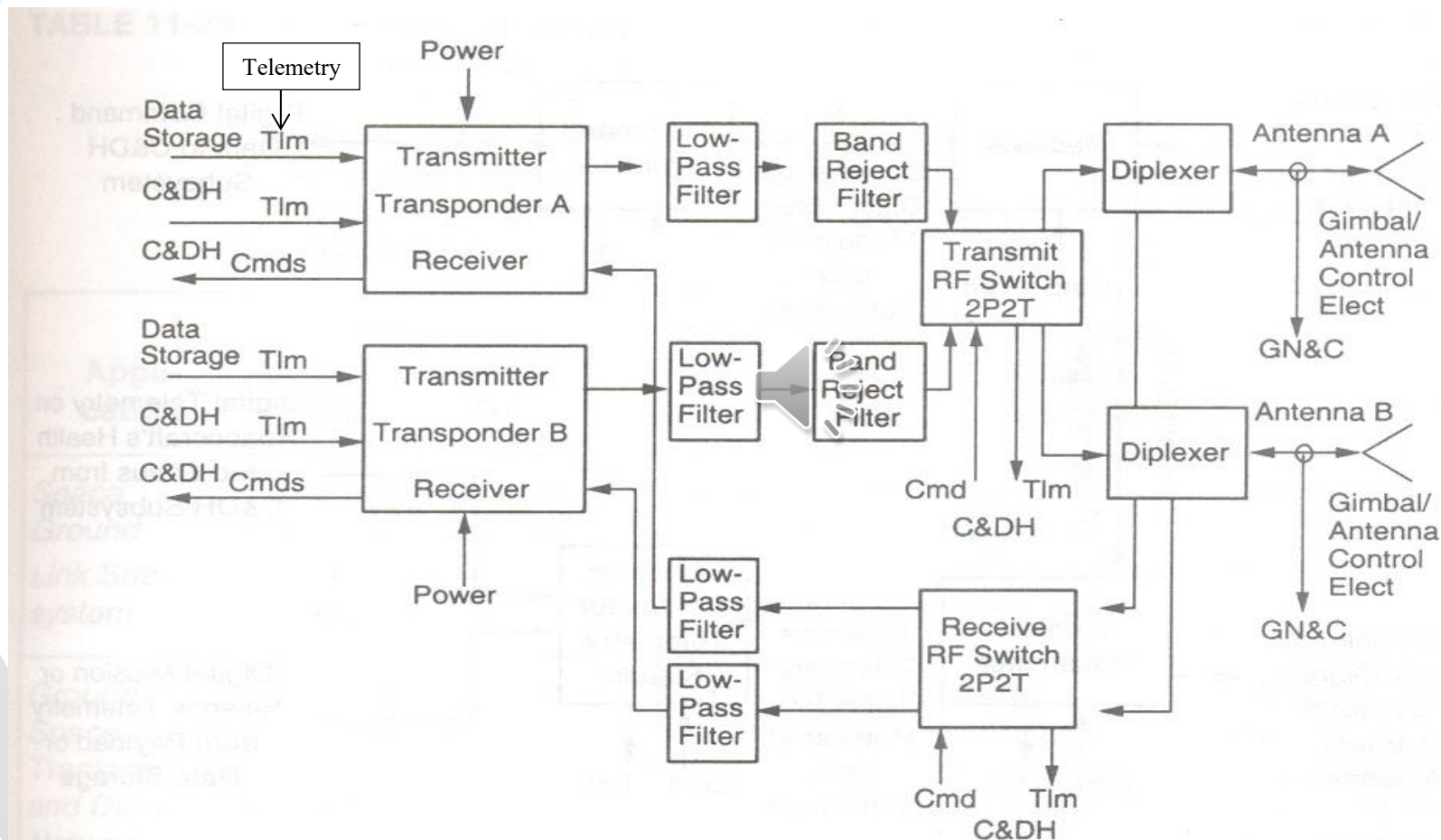


Fig. 11-4. Block Diagram of a Generic TT&C Subsystem. This subsystem has full redundancy: two transponders with parallel transmit and receive signal paths. The diplexer allows the same antenna to be used for transmitting and receiving. The band-reject filter attenuates spurious signals originating from the transmitter at the receiver's center frequency to help the diplexer isolate the receiver from the transmitter.

◆ 11.2.2 Designing the TTC Subsystem (cont'd)

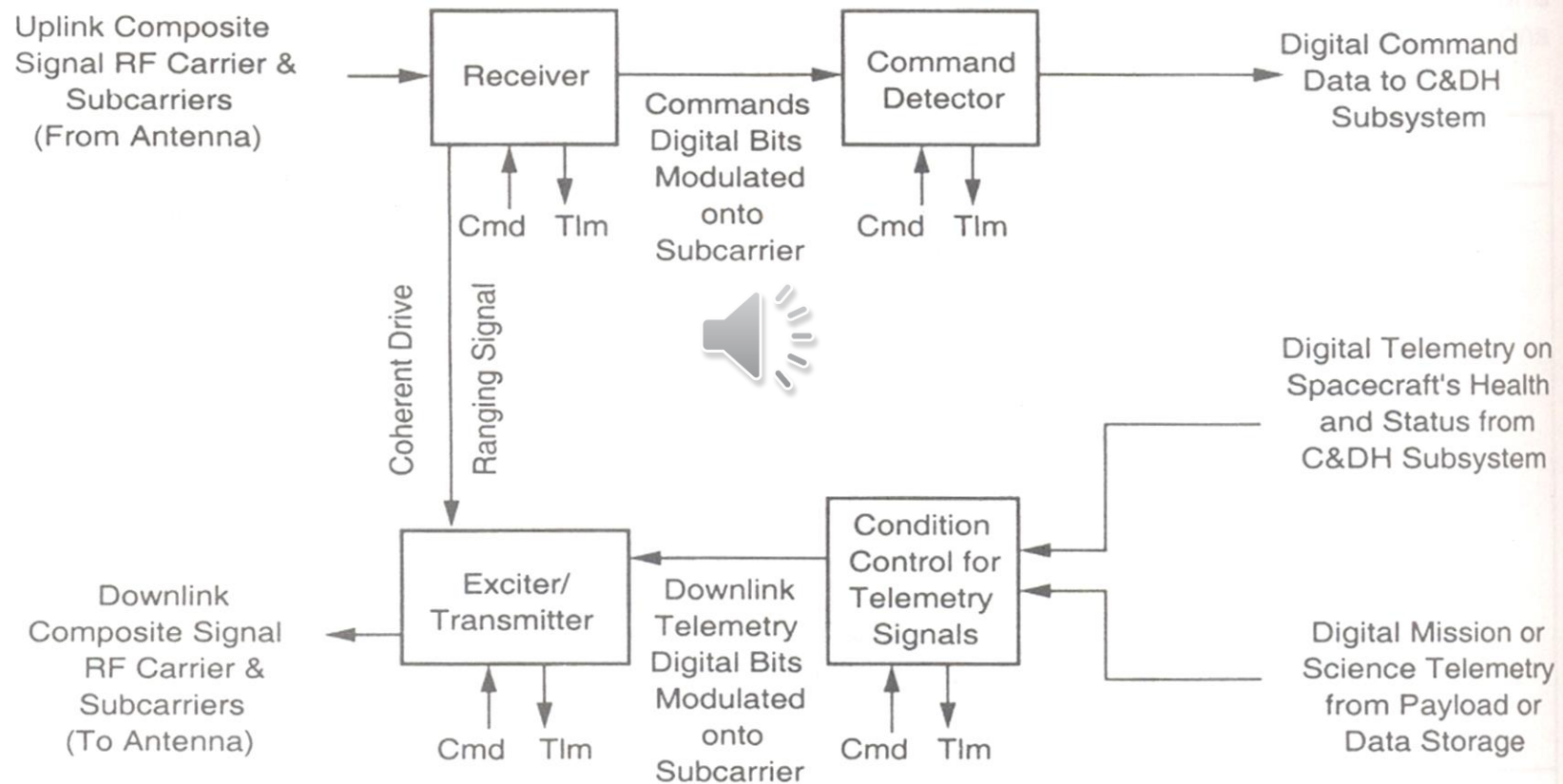


Fig. 11-5. Block Diagram Showing How a Typical Transponder Works. The receiver is isolated from the transmitter except for the ranging signal path and the coherent drive path. The coherent drive comes from the receiver's voltage-controlled oscillator when we want the downlink carrier to be a ratio of the uplink carrier.

◆ 11.2.2 Designing the TTC Subsystem (cont'd)

TABLE 11-25. Attributes of Common Telemetry, Tracking, and Subsystems. This table presents the principal characteristics of two uplink/downlink systems and three cross-link systems. Each system can support various modulation schemes. The antenna characteristics for the uplink/downlink systems contain an Earth-coverage antenna for nominal operations and a hemispherical-coverage antenna for emergency operations. See Sec. 13.3 for modulation descriptions. (Courtesy of TRW)

Appli- cation	Frequency		Modulation		Antenna Charac- teristics	Remarks
	U/L	D/L	U/L	D/L		
Space Ground Link Sub- system	S-band 1.75 to 1.85 GHz	S-band 2.20 to 2.30 GHz	FSK AM PM	PCM PM FM	Earth coverage; Hemispherical coverage	SGLS Standard, refer to TOR-0059 Reissue H
Ground Space Tracking and Data Network	S-band 2.02 to 2.12 GHz	S-band 2.20 to 2.30 GHz	PCM PSK FSK	PCM PSK PM	Earth coverage; Hemispherical coverage	GSTDN is slowly phasing out. JPL-DSN is absorbing some of its assets. (See JPL-DSN-810-5 rev D)

Legend:

FSK = Frequency Shift Keying

AM = Amplitude Modulation

FM = Frequency Modulation

QPSK = Quadrature Phase Shift Keying

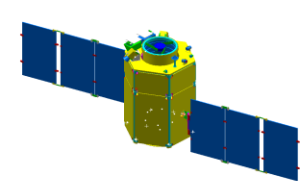
PSK = Phase Shift Keying

PCM = Pulsed-Code Modulation

PM = Phase Modulation

*PQM = Phase Quadrature
Modulation*

PPM = Pulse Position Modulation



◆ 11.2.2 Designing the TTC Subsystem (cont'd)

Appli- cation	Frequency		Modulation		Antenna Charac- teristics	Remarks
	U/L	D/L	U/L	D/L		
<i>Cross-link Within Constel- lation</i>	W-band 60 GHz	W-band 60 GHz	Any	Any	Narrow beam 0.1 deg typical	Modulation, coding, and encryption can be customized
<i>Cross-link to TDRSS</i>	S-band K-band Ku-band	S-band K-band Ku-band	QPSK Spread spectrum	QPSK Spread spectrum	Narrow beam	TDRSS User Standard (See NASA Goddard STDN 101.2 TDRSS Users' Guide)
<i>Optical</i>	IR to UV	IR to UV	PPM/ PCM	PPM/ PCM	Telescope pencil beam	Frequency depends on device

Legend:

FSK = Frequency Shift Keying

AM = Amplitude Modulation

FM = Frequency Modulation

QPSK = Quadrature Phase Shift Keying

PSK = Phase Shift Keying

PCM = Pulsed-Code Modulation

PM = Phase Modulation

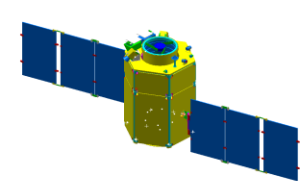
*PQM = Phase Quadrature
Modulation*

PPM = Pulse Position Modulation

◆ 11.2.2 Designing the TTC Subsystem (cont'd)

TABLE 11-26. Typical Parameters for TT&C Subsystems. Two transponders provide redundancy and two hemispherical-coverage antennas offer full coverage. The Ku-band system assumes another (S-band) system applies when the spacecraft deviates from its proper attitude. (Courtesy of TRW and L3 Communications)

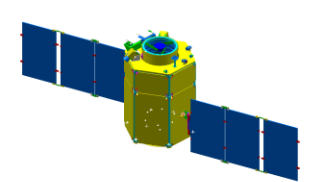
Component	Qty	Mass (kg)		Power (W)	Dimensions (cm)	Remarks
		Each	Total			
Typical X-Band Communication Subsystem						
• Transponder – Receiver – Transmitter	2	3.8	7.6	10.4 35.0 0.0	20 × 22 × 7	Generic X-band transponder – 3-W RF output – Solid-state power amp
• Filters/switch diplexers.	1	1.5	1.5	0.0	10 × 22 × 4	1 set
• Antennas						
– Hemis	2	0.25	0.5	0.0	8.0 dia × 4	Circular Wave Guide
– Parabola	1	9.2	9.2	0.0	150 dia × 70	4-dBi* gain
– Waveguide	1	1.4	1.4	0.0	200 cm long	WR112
TOTAL			20.2	45.4		
Typical S-Band TDRSS User Communication Subsystem						
• Transponder – Receiver – Transmitter	2	6.87	13.74	17.5 40.0	14 × 33 × 7	2nd generation TDRSS user – 12-W RF output – Solid-state power amp
• Filters/switch diplexers	1	2.0	2.0	0.0	15 × 30 × 6	1 set
• Antennas						
– Hemis	2	0.4	0.8	0.0	9.5 dia × 13	Circular Wave Guide
– Parabola	1	9.2	9.2	0.0	150 dia × 70	4-dBi gain
– Turnstile	1	2.3	2.3	0.0	10 dia × 15	Cavity type
– Coax cables	1	0.5	0.5		1.2 dia × 150	1 set
TOTAL			28.54	57.5		



◆ 11.2.2 Designing the TTC Subsystem (cont'd)

Component	Qty	Mass (kg)		Power (W)	Dimensions (cm)	Remarks
		Each	Total			
Typical Ku-Band Communication Subsystem						
• Transponder – Receiver – Transmitter	2	4.45	8.90	4.3 20.0	17 × 34 × 9	Generic Ku-band transponder – 4-W RF output – Solid-state power amp
• Filters/switch diplexers	1	1.2	1.2	0.0	8 × 19 × 4	1 set
• Antennas						
– Earth cover	1	0.5	0.5	0.0	4.0 dia × 2	Earth coverage horn
– Parabola	1	2.0	2.0	0.0	60 dia × 22	Cross-link antenna
– Waveguide	1	0.7	0.7	0.0	125 cm long	
TOTAL			13.3	24.3		

* dBi is the antenna gain relative to an isotropic radiator expressed in decibels (dB). See Sec. 13.3.5



11.3 Command & Data Handling Subsystem

- ◆ C&DH performs two major functions:
 - Receives/Validates/Decodes/Distributes command to other S/C subsystem
 - Gather/Processes/Formats S/C housekeeping and mission data for downlink or use by an onboard computer

- ◆ 11.3.1 Introduction to C&DH
 - Command Handling
 - Consists of a synchronization code, S/C address bits, command messages bits, and error check bits
 - Received commands are validated, prior to execution
 - Validation consists of receiving synchronization code, checking command message length, matching S/C address and fixed bit patterns
 - After validation, the message bit are passed to a decoder for execution
 - Data Handling
 - Combines telemetry from multiple sources
 - Provides it for downlink or interval S/C use.



Command Sources

Command Outputs

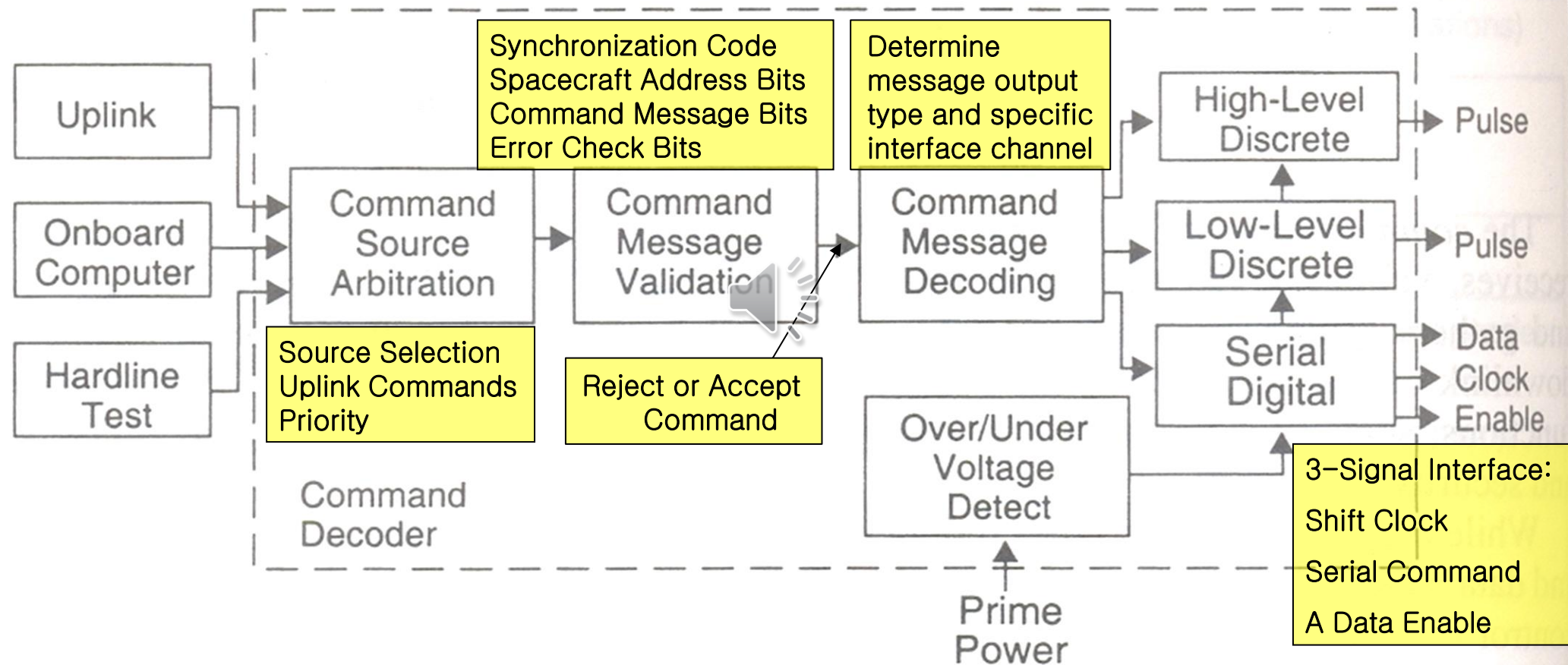


Fig. 11-6. Command Decoder Block Diagram.

11.3 C&DH

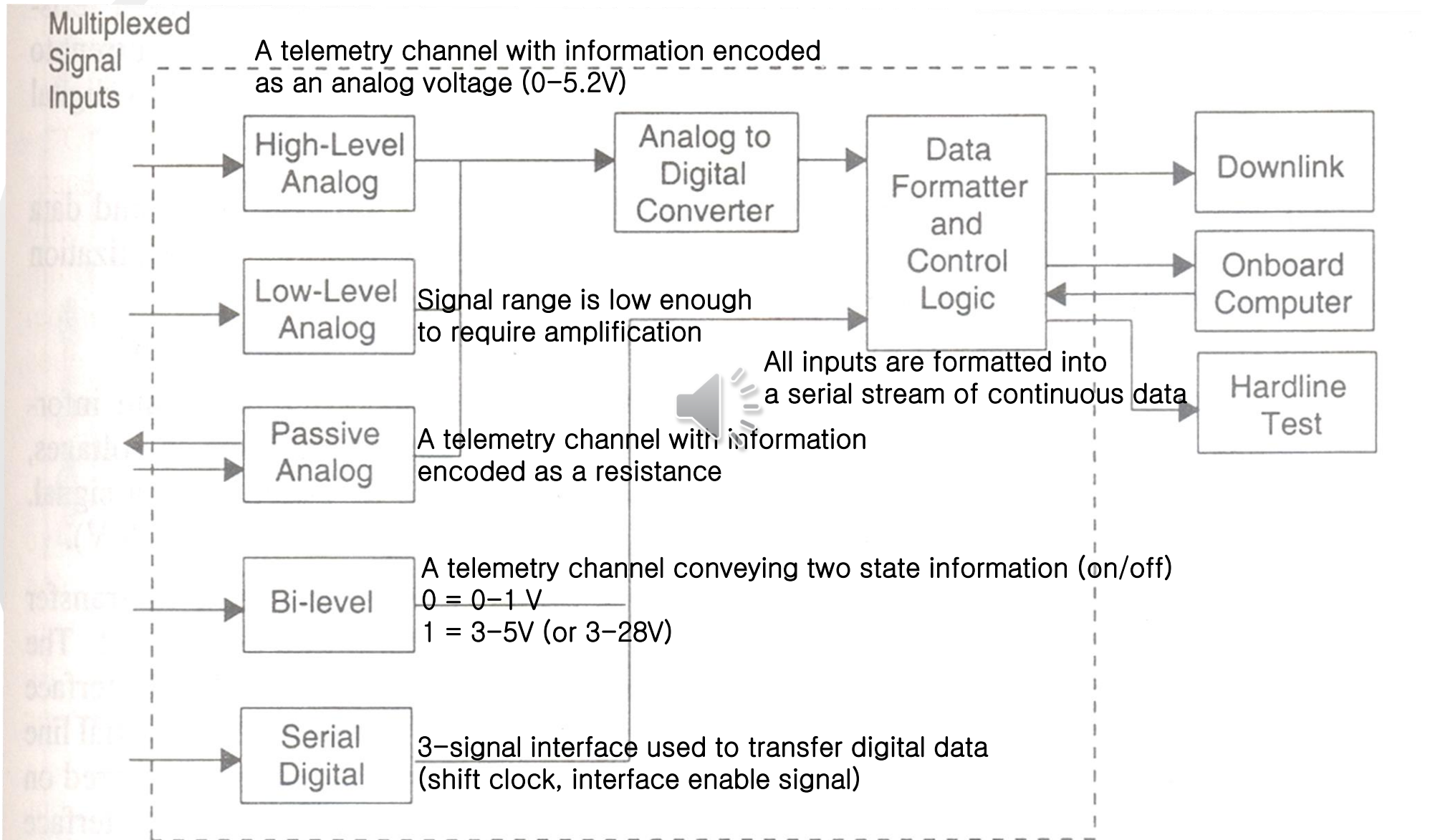


Fig. 11-7. Data Handling Unit Block Diagram.

◆ 11.3.2 C&DH System Sizing Process

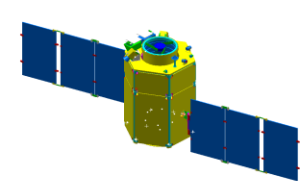
TABLE 11-27. Command and Data Handling System Baseline Process.

Step	Procedure	Issues or Data Needed	FireSat Example	Ref.
1. Identify which functions are to be performed by the C&DH system	Determine or estimate what on the spacecraft must be controlled and what must be monitored to accomplish or support the mission. Determine what tasks are to be performed on board.	Command Processing Is command processing required? If Yes, – What is the command rate? – How many channels? – Is there a computer? – Are stored commands needed? Telemetry Processing Is telemetry processing required? If Yes, – How many channels? – What is the housekeeping data rate? – What is the payload data rate? – Is a computer interface needed? Other Mission time clock needed? Computer watchdog needed? ACS functions needed?	Yes Uncertain < 200 Computer or stored commands, not both Yes < 200 Low 150 Mbps No Yes No Uncertain	Secs. 10.3, 13.2 Secs. 10.3.4, 13.2, 9.5 Sec. 11.1, Chap. 16
2. Identify requirements and constraints	Determine the parameters, driven by aspects of the overall spacecraft design, which impact the C&DH system (the C&DH drivers).	Bus constraints Reliability Satellite Lifetime Radiation Environment Schedule Budget	LightSat bus 0.98 5 years Must be SEU hard 2-year development Low	Chap. 10 Secs. 19.2 8.2
3. Determine complexity of C&DH functions	Use Table 11-28 to estimate the complexity of each function identified in step 1, applying constraints established in step 2.	Evaluate each function separately	See Table 11-28	Table 11-28
4. Determine overall C&DH level of complexity	Collect functions into command and telemetry components and determine composite complexity of each	Are other functions such as the mission time clock combined into command or telemetry (typically telemetry)?	Mission time clock to be included in telemetry component of C&DH (critical to correlate data with time of day)	
5. Estimate size, weight, and power for each component	Apply results of step 4 to Table 11-29 for command and telemetry components	Combined systems (share housing, reduce interface cabling, may share power supply)	See Table 11-29	Table 11-29

◆ 11.3.2 C&DH System Sizing Process (cont'd)

TABLE 11-28. System Complexity Definition. Reliability calculations assume a 5 year mission. Reliability predictions are affected dramatically by mission duration. Simple and typical reliability estimates assume single units. Complex assumes integrated command and data handling.

Requirement or Constraint	System Complexity			FireSat Requirements	FireSat Complexity Selection
	Simple	Typical	Complex		
Processing Commands: CMD rates Computer interface Stored commands Number of channels	50 cmds/s none none < 200 channels	50 cmds/s Computer or stored cmds (not both) 300–500 channels	≥ 50 cmds/s yes not needed > 500 channels	Required ≤ 50 cmds/s Computer or stored commands needed < 200	Simple-typical
Processing of Telemetry Data: TLM rates Housekeeping data Payload data Computer interface Number of channels	500–4kbps none none < 200 channels	4–64 kbps 1–200 kbps none 400–700 channels	64–256 kbps 10 kbps–10 Mbps yes > 500 channels	Required low 150 Mbps not required < 200	Simple Design Driver Implies separate data link for payload
Other: Mission time clock Computer watchdog ACS functions	none none none	included included if OBC none	included included included	Time required for data time- tagging and remote control ACS/autonomy requirement determined by ACS design	Typical–complex
Bus Constraints	Single Unit	Single unit or multiple units	Integrated or distributed	LightSat bus (single unit)	Simple
Reliability-Class B parts Single String Redundant	0.8233 0.9875	0.7610 0.9736	0.6983 0.9496	0.98 desired	Implies redundancy and Class S parts
Reliability-Class S parts Single string Redundant	0.9394 0.9987	0.9083 0.9964	0.8285 0.9829		
Radiation Environment (total dose)	< 2 krad	2–50 krad	50K–1 Mrads	Low (< 2 krad)	Simple
Schedule (in months, after order) Class B parts Class S parts	6–12 9–18	6–12 9–24	9–18 9–24	Assumed to be driven by system design	24-month schedule acceptable



◆ 11.3.2 C&DH System Sizing Process (cont'd)

TABLE 11-29. Parametric Estimation of C&DH Size, Weight, and Power. Peak command power will vary by command type, duration, and load. Nominal command power will be higher if ACS drivers are required.

		Simple	Typical	Complex	FireSat Baseline
Size (cm ³)	Command only	1,500–3,000	2,000–4,000	5,000–6,000	3,000
	Telemetry only	1,500–3,000	4,000–6,000	9,000–10,000	3,000
	Combined systems	2,500–6,000	6,000–9,000	13,000–1,500	
Weight (kg)	Command only	1.5–2.5	1.5–3.0	4.0–5.0	2.5
	Telemetry only	1.5–2.5	2.5–4.0	6.5–7.5	2.5
	Combined systems	2.75–5.5	4.5–6.5	9.5–10.5	
Power (nominal) (W)	Command only	2	2	2	2
	Telemetry only	5–10	10–16	13–20	10
	Combined systems	7–12	13–18	15–25	